

Offline Emergency Communications Server for Raspberry Pi 5

The **Incident Management System** is a self-contained emergency communications server running on a Raspberry Pi 5 (16 GB) in a Pironman MAX 5 enclosure with dual 1 TB NVMe SSDs in RAID 1. It broadcasts its own Wi-Fi network (**EMCOMM-NET**) via an ASUS RT-BE58 Go router and serves a full suite of amateur radio, ICS, and SAR tools to any browser on scene — no internet, no app installation, no configuration required. Everything is accessible at <http://192.168.50.1>.

29 Web pages

3 Dashboard modes

10 Background services

RAID 1 2× 1TB NVMe

■ Amateur Radio Mode

- Net Control Logger — multi-net, FCC auto-fill, ICS-309 export
- Observer Mode — read-only live net view for served agencies
- Repeater Database — RepeaterBook data, band/ARES/mode filters
- NTS Radiogram — formatted ARRL traffic with log
- Callsign Lookup — full FCC database (~800K licensees) offline
- HF Propagation — band conditions, MUF/LUF, solar indices
- Dead Man's Switch — inactivity alert per armed net
- APRS Tactical Map — live Graywolf + YAAC stations on Leaflet

■ Starcom / Public Safety Mode

- Starcom Net Logger — Radio ID / unit-based public safety nets
- SAR Net — quick-launch Search & Rescue net with unit tracking
- Resource Board — 5-state status cycle (Available → Demobilized)
- Facilities Directory — EOC, shelters, hospitals with frequencies
- Weather Net — quick-launch storm spotter check-in net
- Resource Tracking Map — unit placement, status, zone drawing
- Member Roster — Radio IDs, certifications, activation check-in
- ICS Platform access — full incident command from Starcom mode

■ ICS Incident Command — including SAR Activations

- Incident types: Natural Disaster, HazMat, **Search & Rescue**, MCI
- Operations Section — T-card board, resource assignments (ICS-204)
- Logistics Section — comms plan, supply tracking (ICS-205/206/211)
- ICS-213 / ICS-214 / ICS-309 forms — fill, print, archive by incident
- Command Section — objectives, safety, weather, staff (ICS-201–203)
- Planning Section — ICS-209, IAP tracker, resource status table
- Finance/Admin — cost accounting, time tracking, procurement
- Winlink Form Import — parse ICS XML from Winlink Express

■ SAR & Personnel Operations

- SAR incident type — full ICS platform for search and rescue ops
- SAR unit tracking — resource map with ■ Search & Rescue unit type
- Member certifications — ICS-100 through ICS-800, EmComm I/II, CERT
- Search & Rescue Net — Starcom logger net type for SAR radio ops
- Activation tracking — check-ins, walk-ins, deployment log per incident
- Operator access cards — Avery 5371 printable laminated cards

■ Winlink & Digital Comms — Windows Laptop + Pi

- Winlink Express (Windows) — primary Winlink with IC-7300 + VARA HF
- JS8Call (Windows) — HF keyboard-to-keyboard weak-signal digital mode
- Single USB cable from IC-7300 — no interface box required
- Pat Winlink (Pi) — browser-based backup email-over-radio, port 8090
- JS8Call API — open from any EMMCOMM-NET device via dashboard card
- Winlink Form Import — ICS-213/214/309 XML archived to incident

■ Reference & Administration

- Reference Library — upload, search, tag, and serve field documents
- Kiwix Offline Library — WikiMed, Wikipedia, iFixit, Wikivoyage
- Pre-Flight Checklist — GO / CAUTION / NO-GO deployment readiness
- Radio Cheat Sheets — phonetic, Q-codes, prowords, band plan, CTCSS
- Print Center — all forms, reference cards, incident cover sheet
- System Health Monitor — CPU, memory, disk, services, GPS, internet

HOW TO ACCESS

Connect to Wi-Fi **EMCOMM-NET**, open any browser to <http://192.168.50.1> — no app, no login, no internet.

HARDWARE

Raspberry Pi 5 (16 GB) · Pironman MAX 5 · 2× 1 TB NVMe RAID 1 · ASUS RT-BE58 Go Wi-Fi 7 · UniFi Switch Flex 2.5G-5 · Windows laptop (Winlink Express + JS8Call + IC-7300) · Operator stations: Raspberry Pi 500 + Pi Monitor 15.6'